

NOUMAN SAEED BUTT

Machine Learning Developer | Data Science & AI Solutions

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PROFESSIONAL SUMMARY

Machine Learning Developer with hands-on experience building end-to-end ML and deep learning solutions, including predictive healthcare models, explainable AI systems for geospatial image classification, HR analytics, and AI-driven financial planning tools. Skilled in Python, TensorFlow/PyTorch, scikit-learn, and Explainable AI (XAI) techniques, with a strong foundation in data preprocessing, model evaluation, and deployment. Combines technical depth with remote-work experience in administrative and digital operations, bringing strong communication, reliability, and independent project ownership to client engagements.

MACHINE LEARNING PROJECTS

Chronic Kidney Disease Prediction System

- Built a supervised machine learning model to predict chronic kidney disease risk from patient clinical data, supporting early diagnosis and preventive care decisions.
- Performed data cleaning, feature selection, and handling of missing/imbalanced medical data to improve model reliability.
- Trained and evaluated multiple classification algorithms, tuning hyperparameters to optimize accuracy and recall for a healthcare-sensitive use case.

Multiclass Land Cover Recognition Using Explainable Deep Learning Framework

- Designed a deep learning framework using Convolutional Neural Networks (CNNs) to classify Land Use and Land Cover (LULC) types from remote sensing imagery, addressing complex spatial-spectral variation across diverse environmental conditions.
- Integrated Explainable AI (XAI) methods — including Score-CAM, LIME, SHAP, Saliency Maps, Integrated Gradients, SmoothGrad, Gradient x Input, Occlusion, and DeepLIFT — to make model predictions transparent and interpretable for decision-makers.
- Applied preprocessing pipelines and transfer learning to improve model generalization and performance across varied geospatial datasets.
- Validated the model using 5-fold cross-validation (mean accuracy $98.40\% \pm 0.06\%$, Macro-F1 98.38%, MCC 0.984) and achieved 99.65% overall accuracy with a final ensemble model, alongside strong calibration and low failure rate (0.35%).
- Delivered a framework that improves classification accuracy while building trust in automated LULC predictions through visual, explainable outputs.

Employee Attrition Prediction Model

- Developed a machine learning model to predict employee attrition risk using historical HR data, helping organizations identify retention risk factors.
- Conducted exploratory data analysis and feature engineering to surface key drivers of employee turnover.
- Evaluated model performance using precision, recall, and F1-score to ensure actionable, business-relevant predictions.

AI-Powered Financial Planning Assistant

- Built an AI-driven financial planning tool that generates personalized insights and recommendations based on user financial data and goals.
- Integrated data-driven analysis to support budgeting, savings, and financial decision-making with AI-generated insights.
- Designed the system with a focus on clear, actionable output for non-technical end users.

TECHNICAL SKILLS

Languages & Tools: Python, C#, SQL, Firebase

Machine Learning & AI: Supervised & Unsupervised Learning, CNNs, Deep Learning, Explainable AI (XAI: SHAP, LIME, Score-CAM, Grad-CAM techniques), Feature Engineering, Model Evaluation & Tuning

Data & Platforms: Data Preprocessing, Exploratory Data Analysis, Google Workspace, Microsoft Office, Canva

CERTIFICATIONS

- Microsoft — Foundational C# Certification
- Kaggle — Feature Engineering Micro-Certification
- Harvard University — CS50: Introduction to Computer Science